

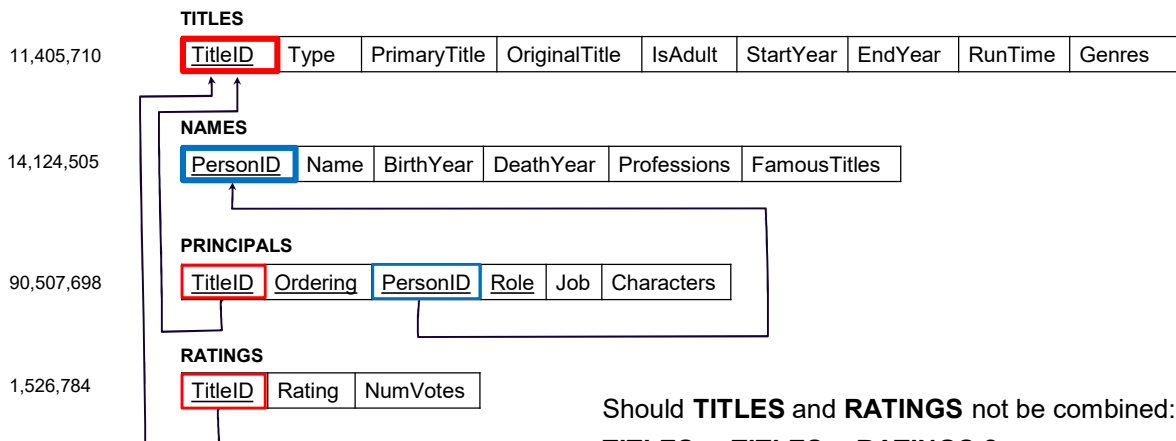
Chapter 9

Functional Dependencies and Normalization

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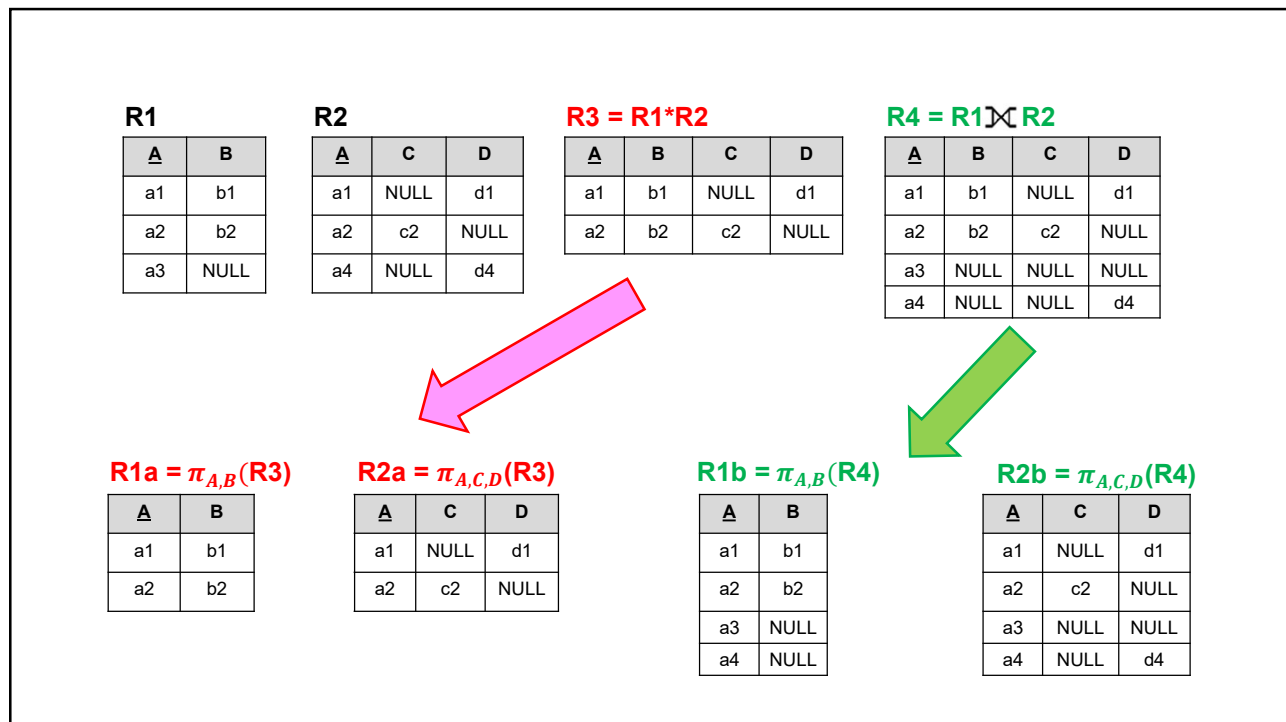


TUPLES

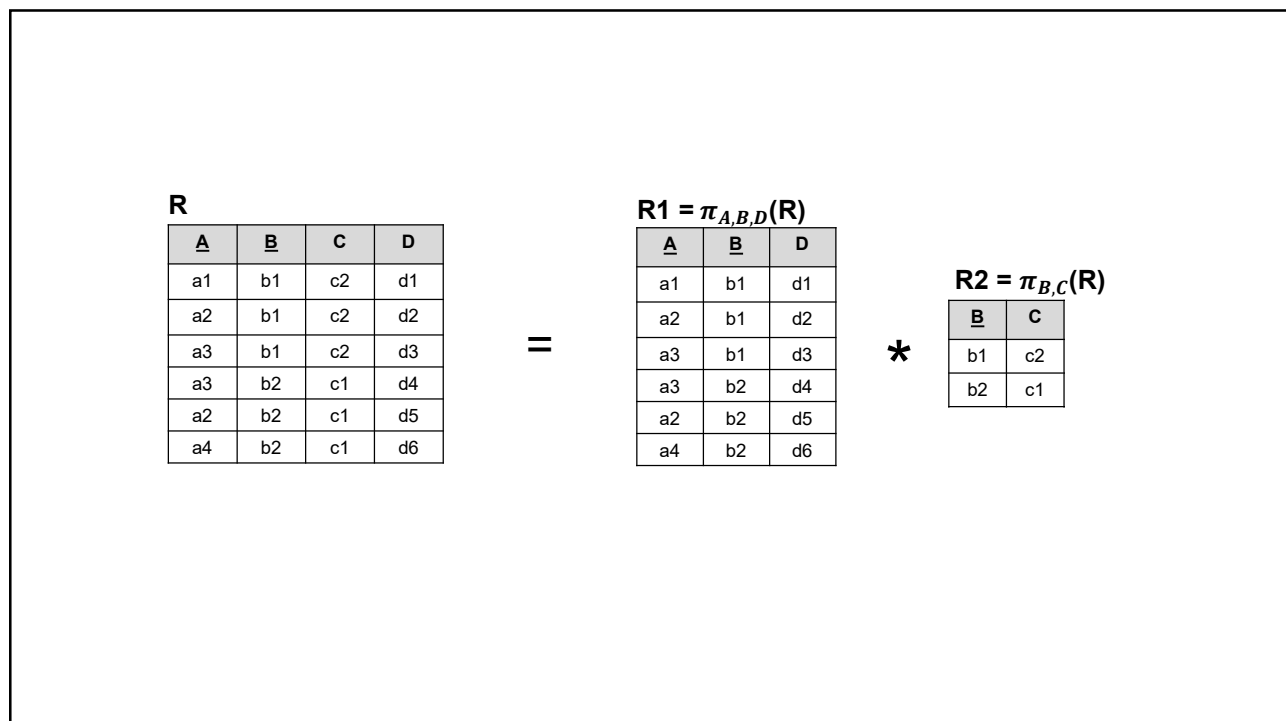


Should **TITLES** and **RATINGS** not be combined:
TITLES ← **TITLES** * **RATINGS** ?

2



3



4

Bad Relational Schema

A relation for a company describing employees working on projects:

EMP_PROJ			Redundancy		Redundancy	
Ssn	Pnumber	Hours	Ename	Pname	Plocation	
123456789	1	32.5	Smith, John B.	ProductX	Bellaire	
123456789	2	7.5	Smith, John B.	ProductY	Sugarland	
666884444	3	40.0	Narayan, Ramesh K.	ProductZ	Houston	
453453453	1	20.0	English, Joyce A.	ProductX	Bellaire	
453453453	2	20.0	English, Joyce A.	ProductY	Sugarland	
333445555	2	10.0	Wong, Franklin T.	ProductY	Sugarland	
333445555	3	10.0	Wong, Franklin T.	ProductZ	Houston	
333445555	10	10.0	Wong, Franklin T.	Computerization	Stafford	
333445555	20	10.0	Wong, Franklin T.	Reorganization	Houston	
999887777	30	30.0	Zelaya, Alicia J.	Newbenefits	Stafford	
999887777	10	10.0	Zelaya, Alicia J.	Computerization	Stafford	
987987987	10	35.0	Jabbar, Ahmad V.	Computerization	Stafford	
987987987	30	5.0	Jabbar, Ahmad V.	Newbenefits	Stafford	
987654321	30	20.0	Wallace, Jennifer S.	Newbenefits	Stafford	
987654321	20	15.0	Wallace, Jennifer S.	Reorganization	Houston	
888665555	20	Null	Borg, James E.	Reorganization	Houston	

Update Anomaly:

Changing the name of project #2 from "ProductY" to "Accounting" requires this update to be made for the many employees working on project #2.

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Other Anomalies

EMP_PROJ					
Ssn	Pnumber	Hours	Ename	Pname	Plocation
123456789	1	32.5	Smith, John B.	ProductX	Bellaire
123456789	2	7.5	Smith, John B.	ProductY	Sugarland
666884444	3	40.0	Narayan, Ramesh K.	ProductZ	Houston
453453453	1	20.0	English, Joyce A.	ProductX	Bellaire
453453453	2	20.0	English, Joyce A.	ProductY	Sugarland
333445555	2	10.0	Wong, Franklin T.	ProductY	Sugarland
333445555	3	10.0	Wong, Franklin T.	ProductZ	Houston
333445555	10	10.0	Wong, Franklin T.	Computerization	Stafford
333445555	20	10.0	Wong, Franklin T.	Reorganization	Houston
999887777	30	30.0	Zelaya, Alicia J.	Newbenefits	Stafford
999887777	10	10.0	Zelaya, Alicia J.	Computerization	Stafford
987987987	10	35.0	Jabbar, Ahmad V.	Computerization	Stafford
987987987	30	5.0	Jabbar, Ahmad V.	Newbenefits	Stafford
987654321	30	20.0	Wallace, Jennifer S.	Newbenefits	Stafford
987654321	20	15.0	Wallace, Jennifer S.	Reorganization	Houston
888665555	20	Null	Borg, James E.	Reorganization	Houston

Insert Anomaly:

- Cannot add a project unless an employee is assigned to it.
- Cannot add an employee unless s/he is assigned to a project.

Delete Anomaly:

- When a project is deleted, it will result in deleting all the employees who work only on that project.
- If an employee is the sole employee on a project, deleting that employee would result in deleting the corresponding project.

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Null Values

- Relations should be designed such that their tuples will have as few NULL values as possible
- Attributes that are NULL frequently could be placed in separate relations

Reasons for nulls:

Attribute not applicable or invalid

Visa_status does not apply to U.S. citizens.

Attribute value unknown

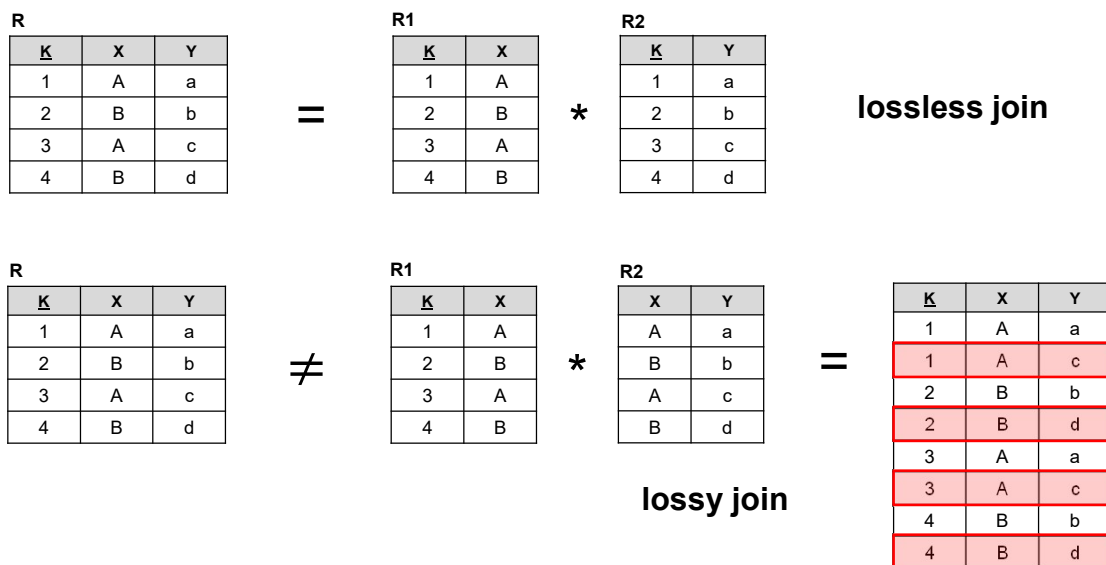
The employee's Date_of_birth may be unknown.

Value does not exist

An employee may not have a Phone_number

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Decomposition



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Naïve Decomposition

EMP_PROJ

Ssn	Pnumber	Hours	Ename	Pname	Plocation
123456789	1	32.5	Smith, John B.	ProductX	Bellaire
123456789	2	7.5	Smith, John B.	ProductY	Sugarland
666884444	3	40.0	Narayan, Ramesh K.	ProductZ	Houston
453453453	1	20.0	English, Joyce A.	ProductX	Bellaire
453453453	2	20.0	English, Joyce A.	ProductY	Sugarland
333445555	2	10.0	Wong, Franklin T.	ProductY	Sugarland
333445555	3	10.0	Wong, Franklin T.	ProductZ	Houston
333445555	10	10.0	Wong, Franklin T.	Computerization	Stafford
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999887777	30	30.0	Zelaya, Alicia J.	Newbenefits	Stafford
999887777	10	10.0	Zelaya, Alicia J.	Computerization	Stafford
987987987	10	35.0	Jabbar, Ahmad V.	Computerization	Stafford
987987987	30	5.0	Jabbar, Ahmad V.	Newbenefits	Stafford
987654321	30	20.0	Wallace, Jennifer S.	Newbenefits	Stafford
987654321	20	15.0	Wallace, Jennifer S.	Reorganization	Houston
888665555	20	Null	Borg, James E.	Reorganization	Houston



EMP_PROJ1

Ssn	Pnumber	Hours	Pname	Plocation
123456789	1	32.5	ProductX	Bellaire
123456789	2	7.5	ProductY	Sugarland
666884444	3	40.0	ProductZ	Houston
453453453	1	20.0	ProductX	Bellaire
453453453	2	20.0	ProductY	Sugarland
333445555	2	10.0	ProductY	Sugarland
333445555	3	10.0	ProductZ	Houston
333445555	10	10.0	Computerization	Stafford
333445555	20	10.0	Reorganization	Houston
999887777	30	30.0	Newbenefits	Stafford
999887777	10	10.0	Computerization	Stafford
987987987	10	35.0	Computerization	Stafford
987987987	30	5.0	Newbenefits	Stafford
987654321	30	20.0	Newbenefits	Stafford
987654321	20	15.0	Reorganization	Houston
888665555	20	NULL	Reorganization	Houston

EMP_LOCS

Ename	Plocation
Smith, John B.	Bellaire
Smith, John B.	Sugarland
Narayan, Ramesh K.	Houston
English, Joyce A.	Bellaire
English, Joyce A.	Sugarland
Wong, Franklin T.	Sugarland
Wong, Franklin T.	Houston
Wong, Franklin T.	Stafford
Zelaya, Alicia J.	Stafford
Jabbar, Ahmad V.	Stafford
Wallace, Jennifer S.	Stafford
Wallace, Jennifer S.	Houston
Borg, James E.	Houston

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Spurious Tuples

EMP_PROJ1 * EMP_LOCS

Ssn	Pnumber	Hours	Pname	Plocation	Ename
123456789	1	32.5	ProductX	Bellaire	Smith, John B.
123456789	1	32.5	ProductX	Bellaire	English, Joyce A.
123456789	2	7.5	ProductY	Sugarland	Smith, John B.
123456789	2	7.5	ProductY	Sugarland	English, Joyce A.
123456789	2	7.5	ProductY	Sugarland	Wong, Franklin T.
666884444	3	40.0	ProductZ	Houston	Narayan, Ramesh K.
666884444	3	40.0	ProductZ	Houston	Wong, Franklin T.
453453453	1	20.0	ProductX	Bellaire	Smith, John B.
453453453	1	20.0	ProductX	Bellaire	English, Joyce A.
453453453	2	20.0	ProductY	Sugarland	Smith, John B.
453453453	2	20.0	ProductY	Sugarland	English, Joyce A.
453453453	2	20.0	ProductY	Sugarland	Wong, Franklin T.
333445555	2	10.0	ProductY	Sugarland	Smith, John B.
333445555	2	10.0	ProductY	Sugarland	English, Joyce A.
333445555	2	10.0	ProductY	Sugarland	Wong, Franklin T.
333445555	3	10.0	ProductZ	Houston	Narayan, Ramesh K.
333445555	3	10.0	ProductZ	Houston	Wong, Franklin T.
333445555	10	10.0	Computerization	Stafford	Wong, Franklin T.
333445555	20	10.0	Reorganization	Houston	Narayan, Ramesh K.
333445555	20	10.0	Reorganization	Houston	Wong, Franklin T.

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Schema Design Guidelines

- Avoid anomalies that may cause redundant work to be done during insertion and modification, and that may cause accidental loss of data during a deletion.
- Avoid waste of storage space due to NULLs and the difficulty of performing selections, aggregation and joins due to NULL values
- Avoid generation of invalid and spurious data during joins on base relations with matched attributes that may not represent a proper (foreign key, primary key) relationship

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Functional Dependencies

A set of attributes X **functionally determines** a set of attributes Y of relation R if the value of X determines a unique value for Y :

$$\underset{\text{determinant}}{X} \rightarrow \underset{\text{dependent}}{Y}$$

i.e. for any two tuples $t_1 \in r(R)$ and $t_2 \in r(R)$:

$$\text{If } t_1[X] = t_2[X], \text{ then } t_1[Y] = t_2[Y]$$

Functional dependencies (FD's)

- specify **formal measures** of the “goodness” of relational designs
- are **constraints** derived from the **meaning** and **interrelationships** of data attributes

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EMPLOYEE

Ename	Ssn	Bdate	Address	Dnumber
Smith, John B.	123456789	1965-01-09	731 Fondren, Houston, TX	5
Wong, Franklin T.	333445555	1955-12-08	638 Voss, Houston, TX	5
Zelaya, Alicia J.	999887777	1968-07-19	3321 Castle, Spring, TX	4
Wallace, Jennifer S.	987654321	1941-06-20	291 Berry, Bellaire, TX	4
Narayan, Ramesh K.	666884444	1962-09-15	975 Fire Oak, Humble, TX	5
English, Joyce A.	453453453	1972-07-31	5631 Rice, Houston, TX	5
Jabbar, Ahmad V.	987987987	1969-03-29	980 Dallas, Houston, TX	4
Borg, James E.	888665555	1937-11-10	450 Stone, Houston, TX	1

SSN → Ename

SSN → Bdate

Ename → Address

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

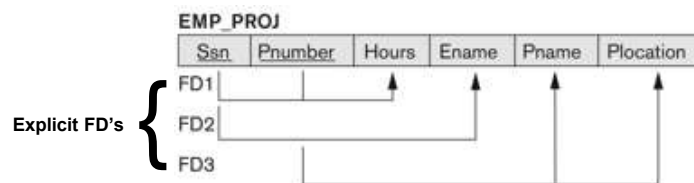
Pnumber → {Pname, Plocation}

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

{Essn, Pno} → Hours

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FD1: {Ssn, Pnumber} → Hours

FD2: Ssn → Ename

FD3: Pnumber → {Pname, Plocation}

Other FD's may be derived from these

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Ruling Out FD's

TEACH

Teacher	Course	Text
Smith	Data Structures	Bartram
Smith	Data Management	Martin
Hall	Compilers	Hoffman
Brown	Data Structures	Horowitz

This FD **may** exist:

Text $\rightarrow$ Course

These FD **cannot** exist:

Teacher $\rightarrow$ Course
Teacher $\rightarrow$ Text
Course $\rightarrow$ Text

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Which FDs May Exist?

A	B	C	D
a1	b1	c1	d1
a1	b2	c2	d2
a2	b2	c2	d3
a3	b3	c4	d3

A $\rightarrow$ B
A $\rightarrow$ C
A $\rightarrow$ D
B $\rightarrow$ C
C $\rightarrow$ B
B $\rightarrow$ D
C $\rightarrow$ D

{A, B} $\rightarrow$ C
{A, B} $\rightarrow$ D
{C, D} $\rightarrow$ B

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Which FDs May Exist?

A	B	C	D	E
a	2	3	4	5
2	a	3	4	5
a	2	3	6	5
a	2	3	6	6

$A \rightarrow B$
 $B \rightarrow A$
 $A \rightarrow BC$
 $DE \rightarrow C$
 $C \rightarrow DE$
 $BC \rightarrow A$

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Armstrong's Inference Rules for FDs

Reflexivity: If $Y \subseteq X$, then $X \rightarrow Y$

Augmentation: If $X \rightarrow Y$, then $XZ \rightarrow YZ$

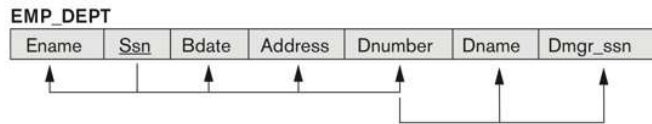
Transitivity: If $X \rightarrow Y$ and $Y \rightarrow Z$, then $X \rightarrow Z$

These form a **sound** and **complete** set of inference rules

Useful to generate new functional dependencies

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Transitivity



$Ssn \rightarrow \{Ename, Bdate, Address, Dnumber\}$

$Dnumber \rightarrow \{Dname, Dmgr_ssn\}$

Therefore:

$Ssn \rightarrow Dname$

$Ssn \rightarrow Dmgr_ssn$

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Other Inference Rules for FDs

Decomposition: If $X \rightarrow YZ$, then $X \rightarrow Y$ and $X \rightarrow Z$

Union: If $X \rightarrow Y$ and $X \rightarrow Z$, then $X \rightarrow YZ$

Pseudotransitivity: If $X \rightarrow Y$ and $WY \rightarrow Z$, then $WX \rightarrow Z$

Because of completeness, these rules can be deduced from Armstrong's inference rules.

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Proof of Decomposition

We are given (a): $X \rightarrow YZ$

By reflexivity: Since $Y \subseteq YZ$, we have (b): $YZ \rightarrow Y$.

By transitivity on (a) and (b): $X \rightarrow Y$

Similarly, we can prove $X \rightarrow Z$

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Using FD to Identify Candidate Keys

Consider a relation $R(A,B,C,D,E)$ with the following dependencies:

$AB \rightarrow C$

$CD \rightarrow E$

$DE \rightarrow B$

What are candidate keys of R ?

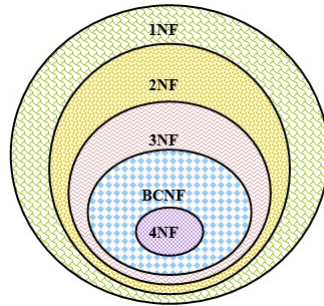
AB (does not imply DE)

ABD ($AB \rightarrow C, CD \rightarrow E$)

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Normalization of Relations

Relationship of Normal Forms



Never leave relations in 1NF or 2NF. Most relations in 3NF are acceptable

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Normalization of Relations

Normalization:

The process of decomposing unsatisfactory “bad” relations by breaking them up into smaller relations.

Normal form:

Condition using keys and FD's of a relation to certify that a relation schema is in a particular “good” form

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First Normal Form (1NF)

A relation is in **first normal form (1NF)** if:

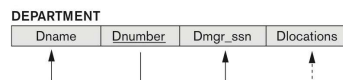
it does not contain:

- multivalued attributes
- composite attributes

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Normalization into 1NF

Schema with multivalued attribute:



Sample state

DEPARTMENT

Dname	Dnumber	Dmgr_ssn	Dlocations
Research	5	333445555	{Bellaire, Sugarland, Houston}
Administration	4	987654321	{Stafford}
Headquarters	1	888665555	{Houston}

1NF version with redundancy

DEPARTMENT

Dname	Dnumber	Dmgr_ssn	Dlocation
Research	5	333445555	Bellaire
Research	5	333445555	Sugarland
Research	5	333445555	Houston
Administration	4	987654321	Stafford
Headquarters	1	888665555	Houston

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Normalization into 1NF

Schema with a composite attribute:

EMP_PROJ		Projs	
Ssn	Ename	Pnumber	Hours

Sample state

EMP_PROJ			
Ssn	Ename	Pnumber	Hours
123456789	Smith, John B.	1	32.5
		2	7.5
666884444	Narayan, Ramesh K.	3	40.0
453453453	English, Joyce A.	1	20.0
		2	20.0
333445555	Wong, Franklin T.	2	10.0
		3	10.0
		10	10.0
		20	10.0
999887777	Zelaya, Alicia J.	30	30.0
		10	10.0
987987987	Jabbar, Ahmad V.	10	35.0
		30	5.0
987654321	Wallace, Jennifer S.	30	20.0
		20	15.0
888665555	Borg, James E.	20	NULL

1NF version by decomposition into two relations by propagating the primary key.

EMP_PROJ1	
Ssn	Ename

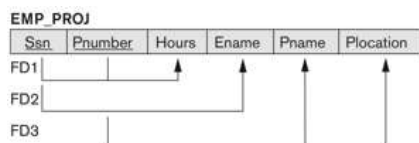
EMP_PROJ2		
Ssn	Pnumber	Hours

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Full Functional Dependency

A functional dependency $X \rightarrow Y$ is a **full FD** if removal of any attribute A from X invalidates the dependency

i.e. for any attribute $A \in X$, $(X - \{A\})$ does *not* functionally determine Y .



$\{Ssn, Pnumber\} \rightarrow Hours$ is a **full FD** since
neither $Ssn \rightarrow Hours$
nor $Pnumber \rightarrow Hours$ hold

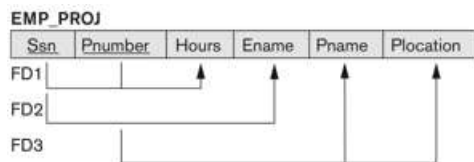
$\{Ssn, Pnumber\} \rightarrow Ename$ is **not** a full FD since
 $Ssn \rightarrow Ename$ also holds
(even though Ssn alone is not a key)

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Partial Functional Dependency

A functional dependency $X \rightarrow Y$ is a **partial FD** if some attribute $A \in X$ can be removed from X without invalidating the dependency.

i.e. for some $A \in X, (X - \{A\}) \rightarrow Y$.



$\{Ssn, Pnumber\} \rightarrow Ename$ is a **partial FD** since

$Ssn \rightarrow Ename$ also holds

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Prime Attributes

A **prime attribute** is a member of **some** candidate key

$\{Ssn\}$

A **nonprime attribute** is not a prime attribute — i.e. not a member of **any** candidate key.

$\{Ename\}$

EMP_PROJ

Ssn	Pnumber	Hours	Ename	Pname	Plocation
123456789	1	32.5	Smith, John B.	ProductX	Bellaire
123456789	2	7.5	Smith, John B.	ProductY	Sugarland
666884444	3	40.0	Narayan, Ramesh K.	ProductZ	Houston
453453453	1	20.0	English, Joyce A.	ProductX	Bellaire
453453453	2	20.0	English, Joyce A.	ProductY	Sugarland
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999887777	10	10.0	Zelaya, Alicia J.	Computerization	Stafford
987987987	10	35.0	Jabbar, Ahmad V.	Computerization	Stafford
987987987	30	5.0	Jabbar, Ahmad V.	Newbenefits	Stafford
987654321	30	20.0	Wallace, Jennifer S.	Newbenefits	Stafford
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888665555	20	Null	Borg, James E.	Reorganization	Houston

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Second Normal Form (2NF)

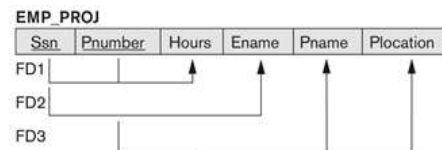
A relational schema R is in **second normal form (2NF)** if

1. It is in 1NF
2. Every **nonprime** attribute $A \in R$ is **fully** functionally dependent on the key (no partial dependencies).

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2NF

Key: {Ssn, Pnumber}
Prime attributes: {Ssn, Pnumber}
Nonprime attributes: {Hours, Ename, Pname, Plocation}



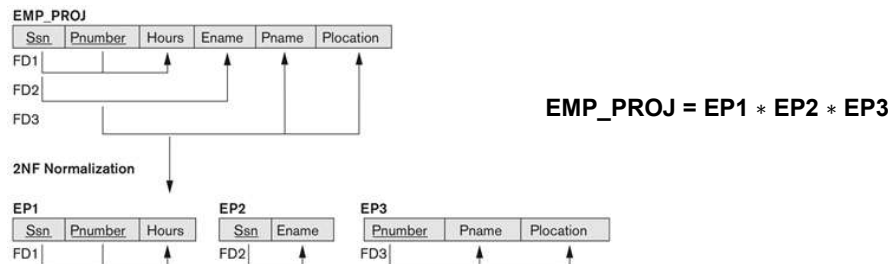
$\{Ssn, Pnumber\} \rightarrow Hours$ is a **full FD**
 $\{Ssn, Pnumber\} \rightarrow Ename$ is not a **full FD** since
SSN $\rightarrow$ Ename also holds
 $\{Ssn, Pnumber\} \rightarrow \{Pname, Plocation\}$ is not a **full FD** since
Pnumber $\rightarrow \{Pname, Plocation\}$ also holds

THUS: EMP_PROJ is not in 2NF !

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Normalizing into 2NF

Decompose into a set of relations in which nonprime attributes are associated only with the part of the key on which they are **fully** functionally dependent.



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2NF for Multiple Keys

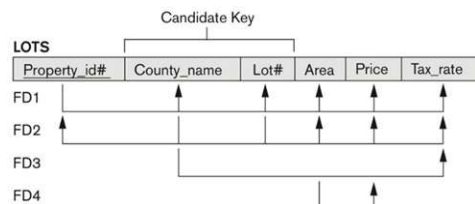
A relation schema R is in **second normal form (2NF)** if:

1. It is in 1NF
2. Every nonprime attribute $A \in R$ is fully functionally dependent on **every candidate key** of R (no partial dependencies).

Prime attributes: {Property_id#, County_name, Lot#}

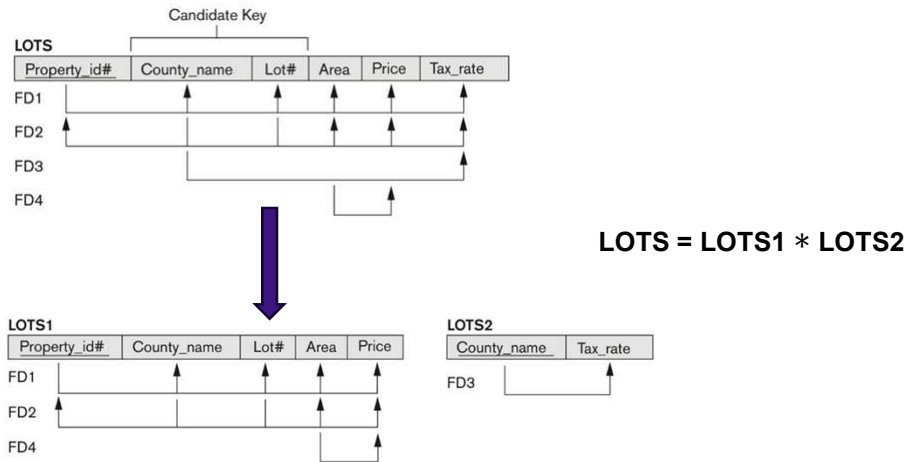
Non_prime attributes: {Area, Price, Tax_rate}

FD3: County_name → Tax_rate violates 2NF



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2NF for Multiple Keys



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Third Normal Form (3NF)

- A relation schema R is in **third normal form (3NF)** if
1. It is in 2NF
 2. No nonprime attribute $A \in R$ is **transitively** dependent on the key

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3NF



Key: {Ssn}
 Prime attributes: {Ssn}
 Nonprime attributes: {Ename, Bdate, Address, Dnumber, Dname, Dmgr_ssn}

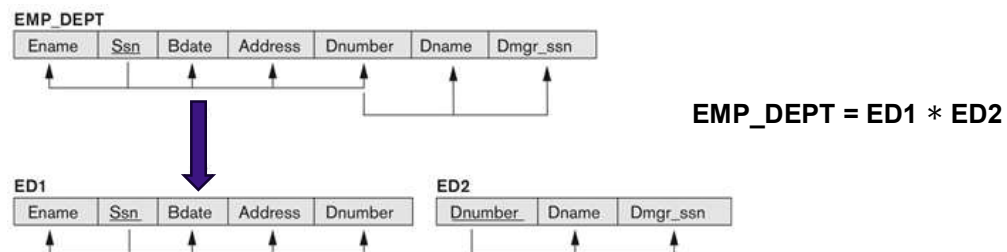
$Ssn \rightarrow Dnumber \rightarrow Dname$ and
 $Ssn \rightarrow Dnumber \rightarrow Dmgr_ssn$ holds

THUS: EMP_DEPT is not in 3NF !

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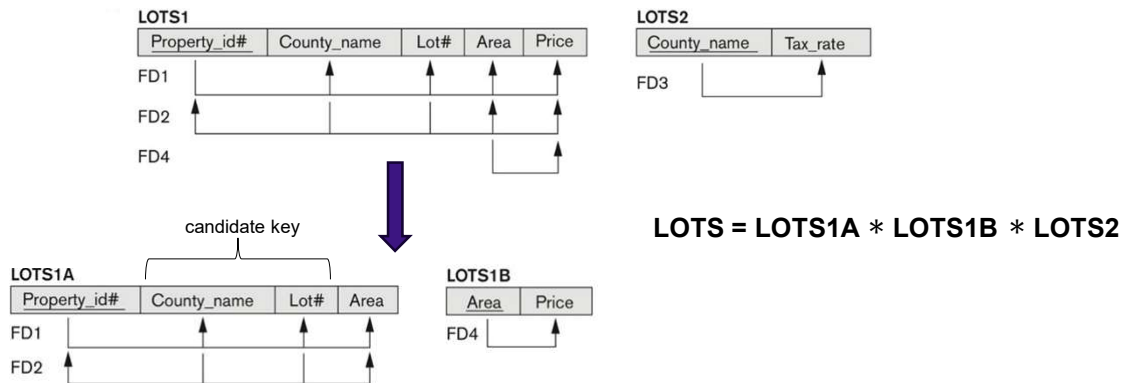
Normalizing into 3NF

Decompose into a set of relations in which nonprime attributes are members **only** with the attributes on which they are functionally dependent (which become candidate keys).



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Normalization into 3NF



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General Definition of 3NF

A relation schema R is in **third normal form (3NF)** if every non-prime attribute of R is both:

- (a) **fully** functionally dependent on **every** key of R
- (b) non-transitively dependent on **every** key of R

Note that this still allows an attribute to be dependent on another **non-prime** attribute.

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3NF:

Every non-prime attribute must provide a fact about the **key**, the **whole** key, and **nothing** but the key

... so help me Codd.

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Example

Consider a relation $R(A,B,C,D,E,F,G,H,I,J)$ with the following functional dependencies:

$AB \rightarrow C$

$A \rightarrow DE$

$B \rightarrow F$

$F \rightarrow GH$

$D \rightarrow IJ$

What are candidate keys for R ? These are attributes whose *functional closure* covers all attributes.

$AB^+ = ABCDEFGHIJ$

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Example

Normalize **R** into 2NF.

Key is AB. FD of prime attributes are:

$A^+ = ADEIJ$, so $A \rightarrow DEIJ$

$B^+ = BFGH$, so $B \rightarrow FGH$

R(A,B,C,D,E,F,G,H,I,J)

$AB \rightarrow C$

$A \rightarrow DE$

$B \rightarrow F$

$F \rightarrow GH$

$D \rightarrow IJ$

Remove the attributes that are functionally dependent on part of the key (A or B) from R and place them in separate relations R1 and R2, along with the part of the key they depend on (A or B):

The key also stays in the remainder of the original relation R3:

R1(A,D,E,I,J) **R2**(B,F,G H) **R3**(A,B,C)

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Example

Continue to normalize **R** into 3NF.

R1 has the transitive dependency $A \rightarrow D \rightarrow IJ$, so remove IJ from R1 into R11 along with D that they are dependent on. The remaining attributes are kept in R12.

R11(D,I,J) **R12**(A,D,E)

R2 is similarly decomposed into R21 and R22 based on the transitive dependency $B \rightarrow F \rightarrow GH$:

R21(E,G,H) **R22**(B,F)

R1(A,D,E,I,J)

R2(B,F,G H)

R3(A,B,C)

$AB \rightarrow C$

$A \rightarrow DE$

$B \rightarrow F$

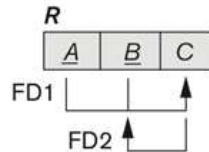
$F \rightarrow GH$

$D \rightarrow IJ$

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BCNF (Boyce-Codd Normal Form)

A relational schema R is in **Boyce-Codd Normal Form (BCNF)** if whenever $X \rightarrow A$ holds in R , then X is a **superkey** of R



R is in 3NF, but not in BCNF due to $FD2: C \rightarrow B$

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TEACH is in 3NF but not in BCNF

TEACH

<u>Student</u>	<u>Course</u>	Instructor
Narayan	Database	Mark
Smith	Database	Navathe
Smith	Operating Systems	Ammar
Smith	Theory	Schulman
Wallace	Database	Mark
Wallace	Operating Systems	Ahamad
Wong	Database	Omiecinski
Zelaya	Database	Navathe
Narayan	Operating Systems	Ammar

FD1: $\{Student, Course\} \rightarrow Instructor$
(each student, for a given subject, is taught by exactly one instructor)

FD2: $Instructor \rightarrow Course$
(each instructor teaches exactly one course)

Redundancy:

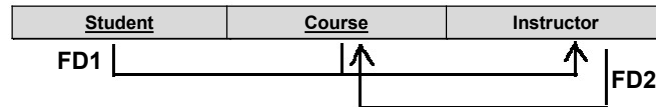
Ammar teaches Operating Systems
Mark teaches Database

appear twice.

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Normalization into BCNF

TEACH



FD1: {Student, Course} → Instructor

FD2: Instructor → Course

Decompose **TEACH** to:

TEACH1(Instructor, Course) and **TEACH2(Instructor, Student)**

The decomposition loses FD1

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TEACH = TEACH1 * TEACH2
no redundancy

TEACH

<u>Student</u>	<u>Course</u>	<u>Instructor</u>
Narayan	Database	Mark
Smith	Database	Navathe
Smith	Operating Systems	Ammar
Smith	Theory	Schulman
Wallace	Database	Mark
Wallace	Operating Systems	Ahamad
Wong	Database	Omiecinski
Zelaya	Database	Navathe
Narayan	Operating Systems	Ammar



TEACH1

<u>Course</u>	<u>Instructor</u>
Database	Mark
Database	Navathe
Operating Systems	Ammar
Theory	Schulman
Operating Systems	Ahamad
Database	Omiecinski

TEACH2

<u>Instructor</u>	<u>Student</u>
Mark	Narayan
Navathe	Smith
Ammar	Smith
Schulman	Smith
Mark	Wallace
Ahamad	Wallace
Omiecinski	Wong
Navathe	Zelaya
Ammar	Narayan

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Normalization into BCNF

To normalize a relation into BCNF:

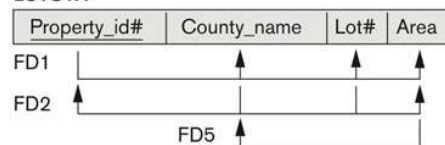
1. List all the determinants.
2. See if each determinant can act as a (candidate) key.
3. For any determinant that is not a candidate key, create a new relation from the FD and remove, while retaining the determinant.

Each step is lossless.

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BCNF Example

LOTS1A



BCNF Normalization

LOTS1AX

Property_id#	Area	Lot#
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LOTS1AY

Area	County_name
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LOTS1A = LOTS1AX * LOTS1AY

BCNF normalization of LOTS1A with FD2 lost in the decomposition.

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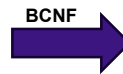
BCNF Example

FUNDS

<u>FundID</u>	<u>InvestmentType</u>	<u>Manager</u>
99	Common Stock	Smith
99	Municipal Bonds	Jones
33	Common Stock	Green
22	Growth Stocks	Brown
11	Common Stock	Smith

FD1: FundID, InvestmentType → Manager
FD2: FundID, Manager → InvestmentType
FD3: Manager → InvestmentType

R1(FundID, Manager) R2(InvestmentType, Manager)



R1

<u>Manager</u>	<u>FundID</u>
Jones	99
Green	33
Brown	22
Smith	11
Smith	99

R2

<u>InvestmentType</u>	<u>Manager</u>
Common Stock	Smith
Municipal Bonds	Jones
Common Stock	Green
Growth Stocks	Brown